Business Intelligence - Program Elective - 1

Course code: PEIT 3101C

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Image source

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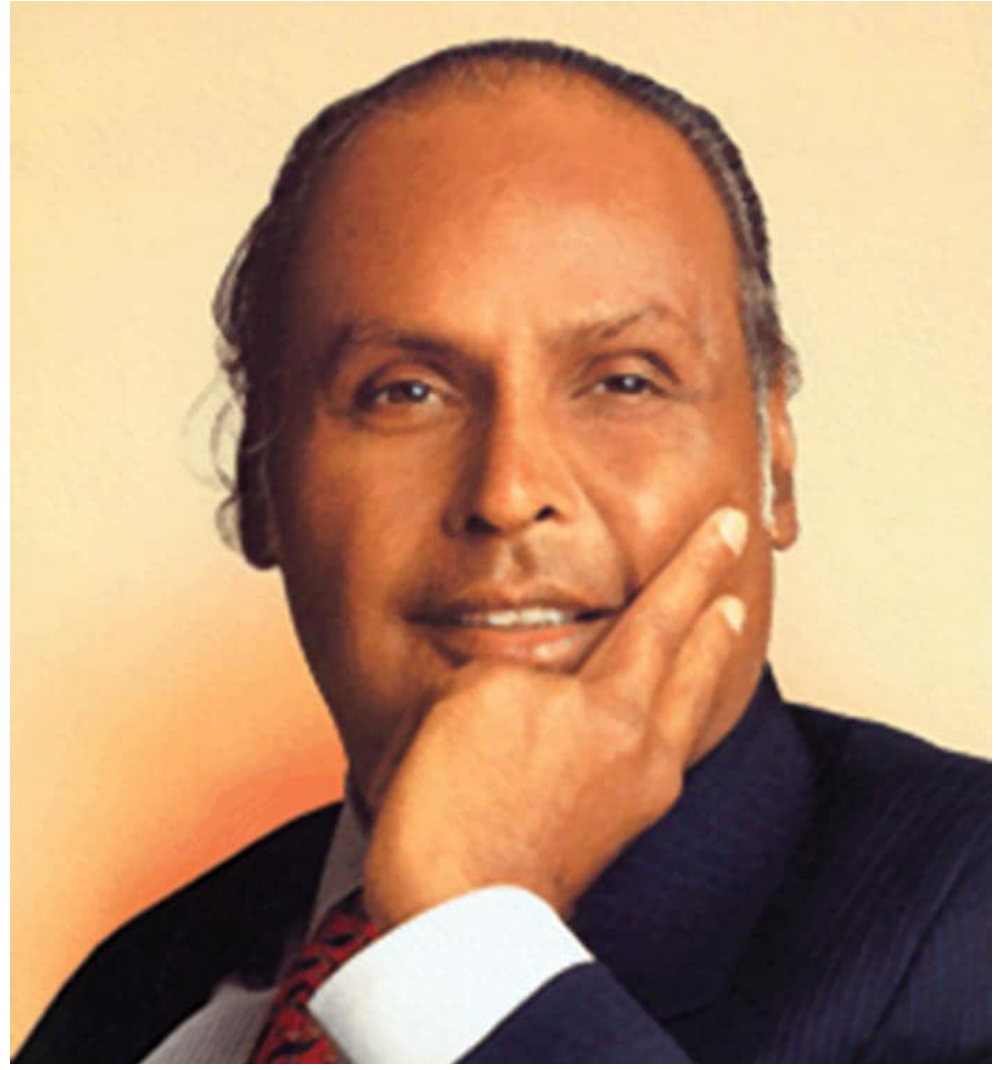
Description

“Broadband will no longer be a luxury”

Reliance Jio Infocomm Ltd is an Indian private sector telecommunication company and a subsidiary of Jio platforms, founded by Mukesh Ambani. It is a brand name under the Reliance Industries Ltd umbrella and claims to be the world’s largest mobile data network.

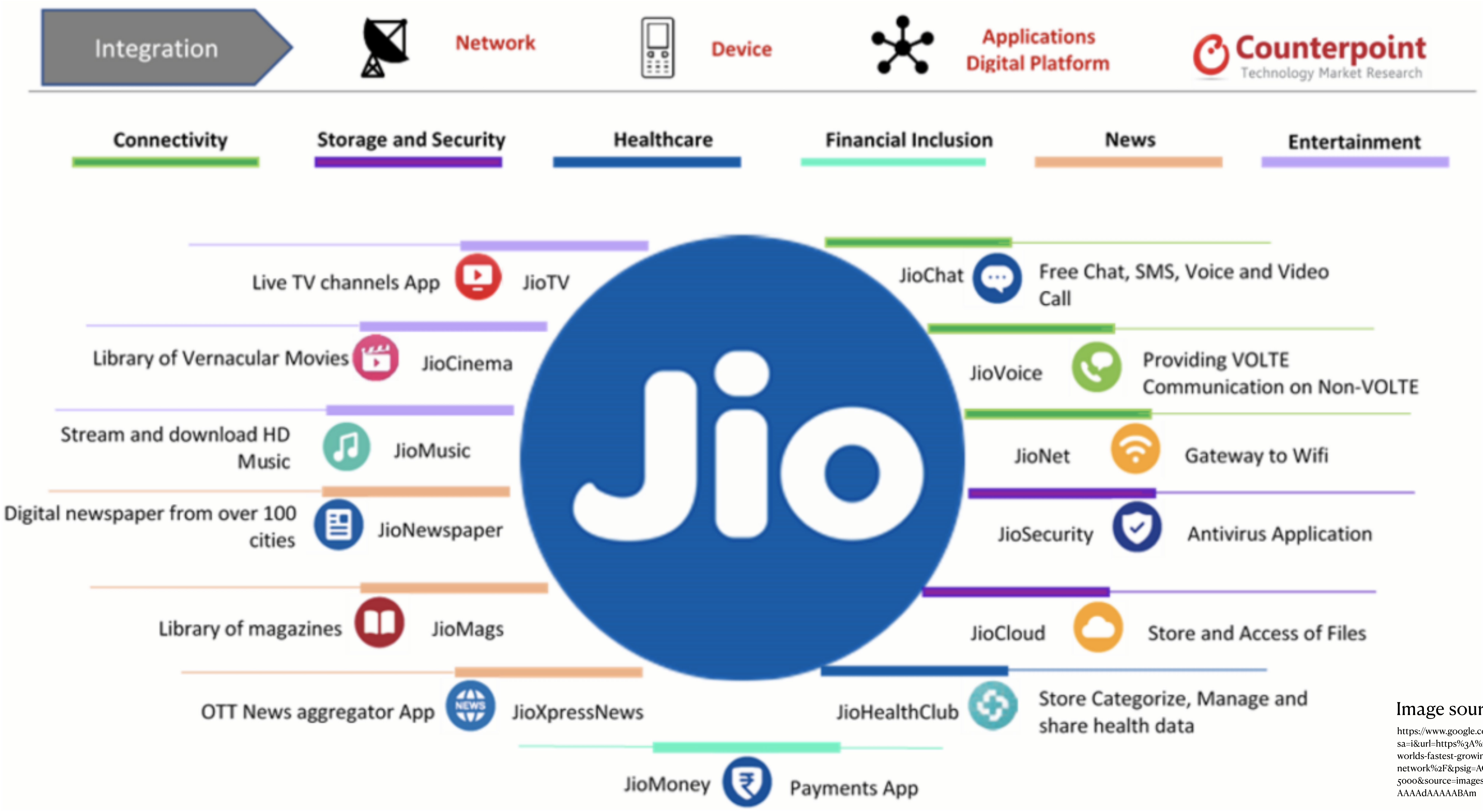


Target audience: Reliance Jio is committed to providing the best and cheapest Internet data services, so their target customers are the **ones having SmartPhone**. They also created a demand for 4G enabled smartphones as their services can only be availed using that.

Their target is not only smartphone users but also the **industries** which require digital technology, bridging the gap between customer merchants and final consumers.

> **“Between my past, the present and the future, there is one common factor: Relationship and Trust. This is the foundation of our growth.”**
>
> \- Dhirubhai Ambani

Products & Services



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https://www.google.co.in/url?sa=i&url=https%3A%2F%2Fwww.kaiostech.com%2Freliance-jio-became-worlds-fastest-growing-mobile-network%2F&psig=AOvVaw23uMxVxp1q4BDhiVctonRK&ust=1634485620315000&source=images&cd=vfe&ved=0CAsQjRxqFwoTCOCj2sqjz_MCFQAAAAAdAAAAABAm

Business Strategy of Reliance Jio

The **AARRR** strategy is one of the biggest reasons for the success of Reliance Jio

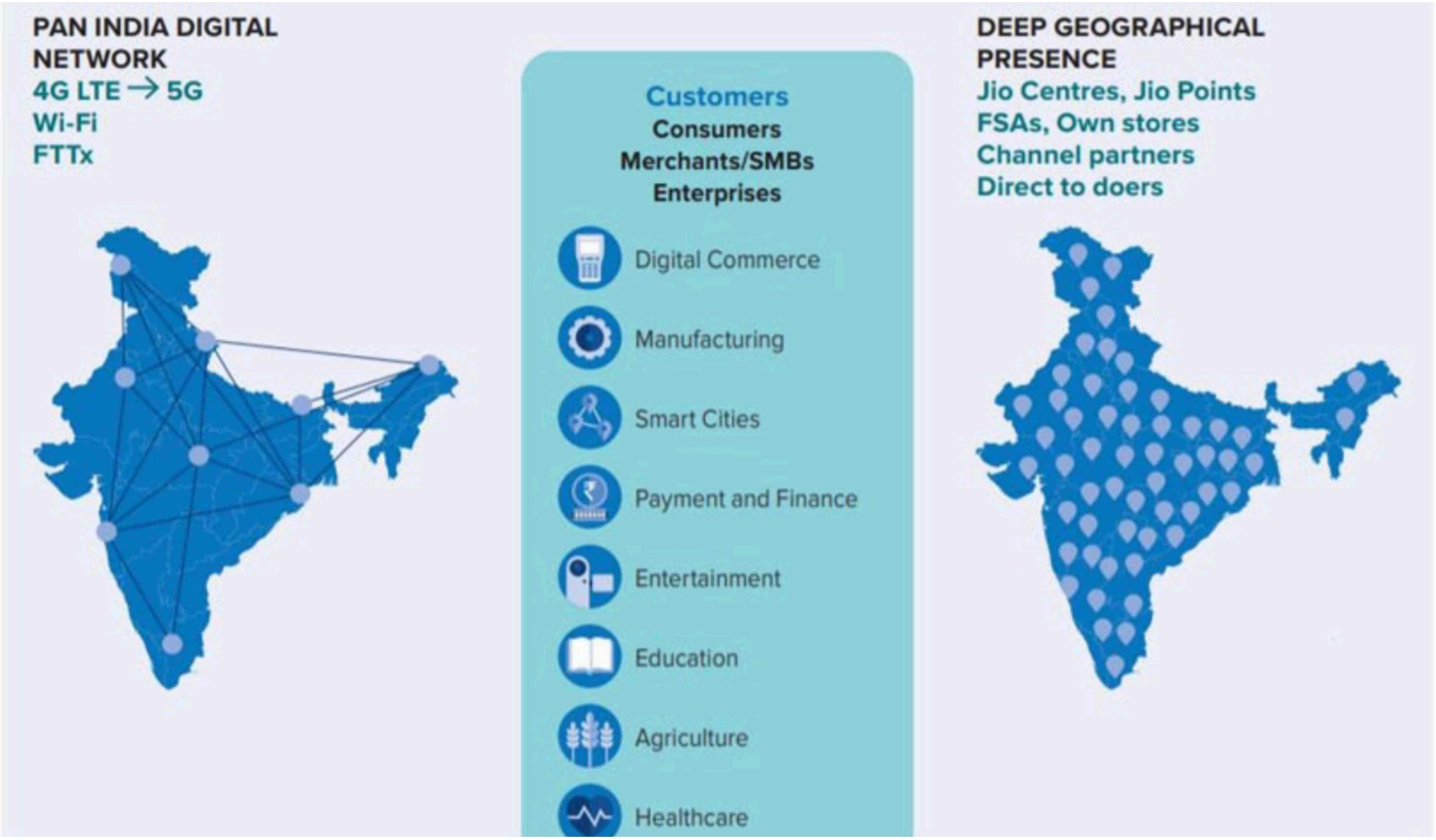
Acquisition
People visit your website

Activation
They have a great first time experience

Retention
Users come back

Referral
Users invite others

Revenue
Users buy your products



Requirement Analysis

User requirements and how they are fulfilled:

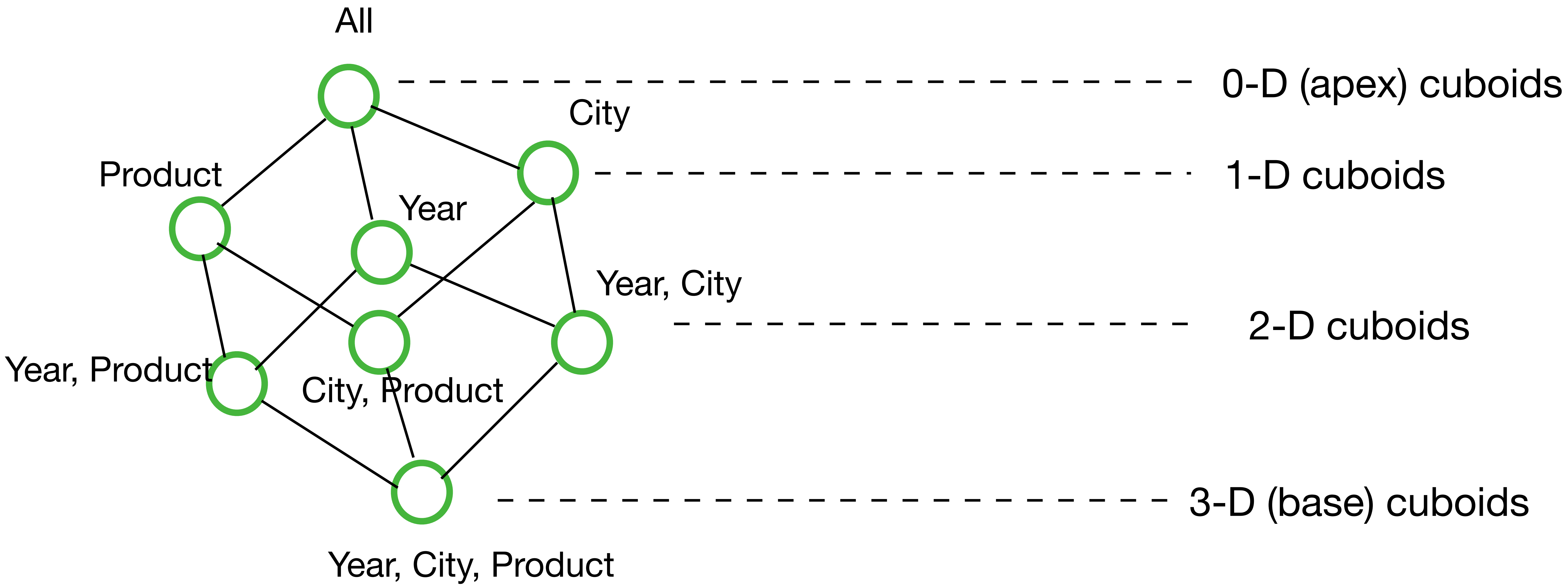
- **Speed** - Number 1 requirement of telecomm users/customers is the speed of the network. Whenever we talk about network, speed is the first thing we compare. Jio was able to capture this main aspect & because they invested much into it, they were able to capture a large market base
- **Value for Money** - There should be value for the price that customers pay, the cost of a telecomm company's products & services. Reliance Jio has its plans and products ranging from high-end to affordable services and this is how their customer segment grew
- **More for Less** - In fact this is also one of the reasons why Jio created a revolution for networking in India. Since Jio offers more benefits for a lesser price than it's competitors, the business grows drastically. "Broadband should no longer be a luxury, it is a necessity" and they strive to live up to it
- **Roaming charges deduction** - Removal of roaming charges across India has lifted a huge burden off the customers travelling the length and breadth of the country
- **Free voice calling to all Jio users** - this is an added benefit which customers have and as result switch to Jio
- **LYF: affordable to all** - RJIL introduces a new smartphone brand LYF which starts at about ₹3000 and operates in 4G. So smartphones and 4G, once considered a luxury, are available to a large number of audience across the country. The growth of their business lies in communication services being afforded by the classes as well as the masses

Grain of Fact Table

The **Grain** or **Granularity** of a fact table refers to the **level of detail** of each row in the fact table. It represents the most atomic level by which the facts may be defined.

For this report I have chosen the dimensions as products, year, city to perform operations hence the grain of the fact table would be: <u>**No. of products sold in a city in a year**</u>.

Data Cube:



Dimension Tables & their Attributes

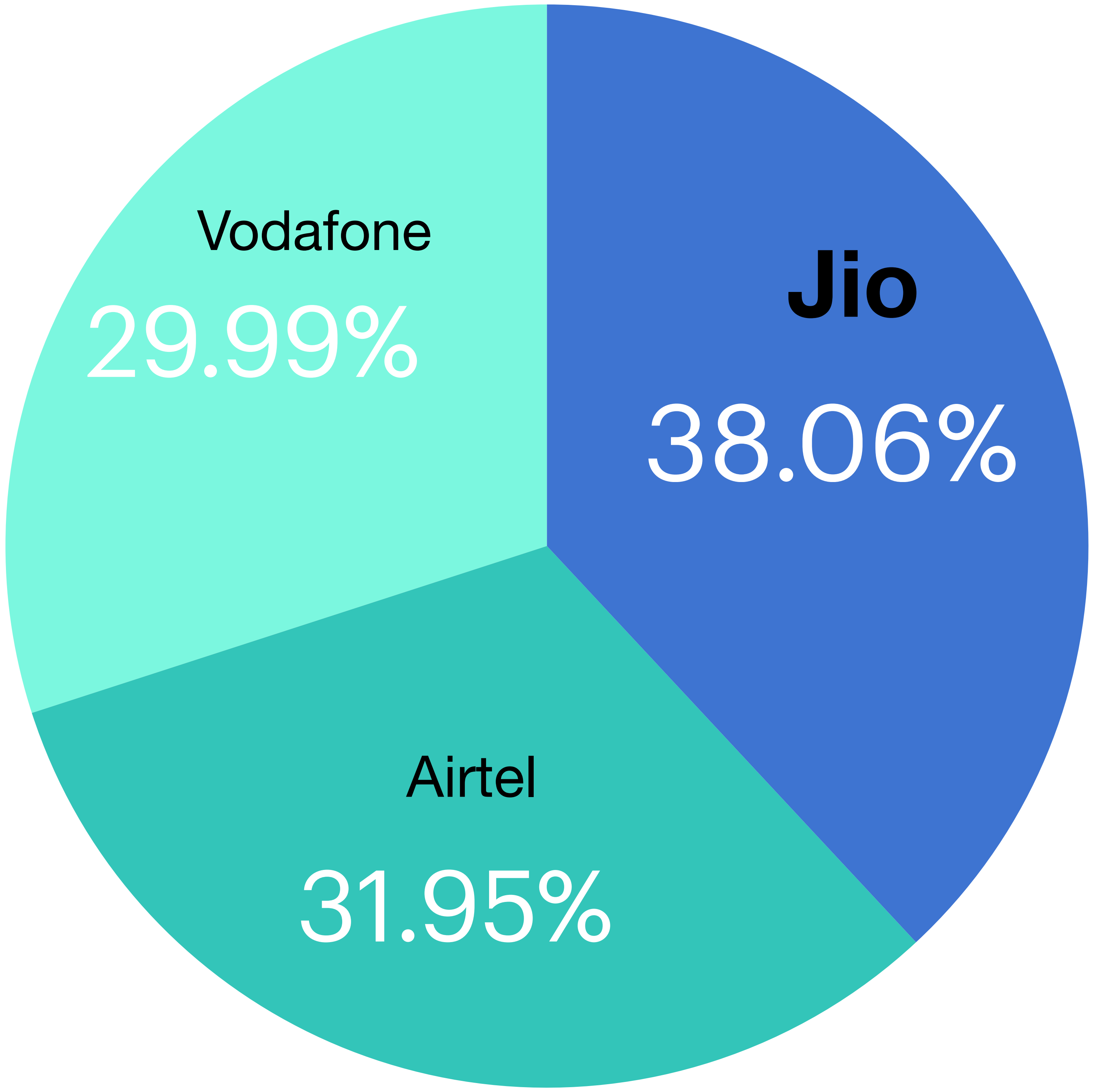
- **Jio_fact** - subscriber_id, store_id, location_id, product_id, plan_id, aadhar_no, retailer_id, employee_id, avg_sales, no_of_subscribers, no_of_retailstores, no_of_products
- **Store_fact** - store_id, retailer_id, location_id, product_id
- **Subscriber_dim** - subscriber_id, sub_name, aadhar_no, sub_products, location_id, age
- **Store_dim** - store_id, store_name, location_id
- **Retailer_dim** - retailer_id, retailer_name, store_id, location_id
- **Employee_dim** - employee_id, employee_name, store_id
- **Location_dim** - location_id, district, city, state, country
- **Product_dim** - product_id, product_name (JioSaavn, JioCinema, JioGigaFiber, JioFi 4G, etc)
- **Plan_dim** - plan_id, plan_name, plan_cost, plan_time
- **Sim_dim** - aadhar_no, sim_no, plan_id

Key Performance Indicators (KPI’s)

KPI’s are the parameters which measure a business’s success based on a particular activity and enable the team to make smart business decisions based on them.

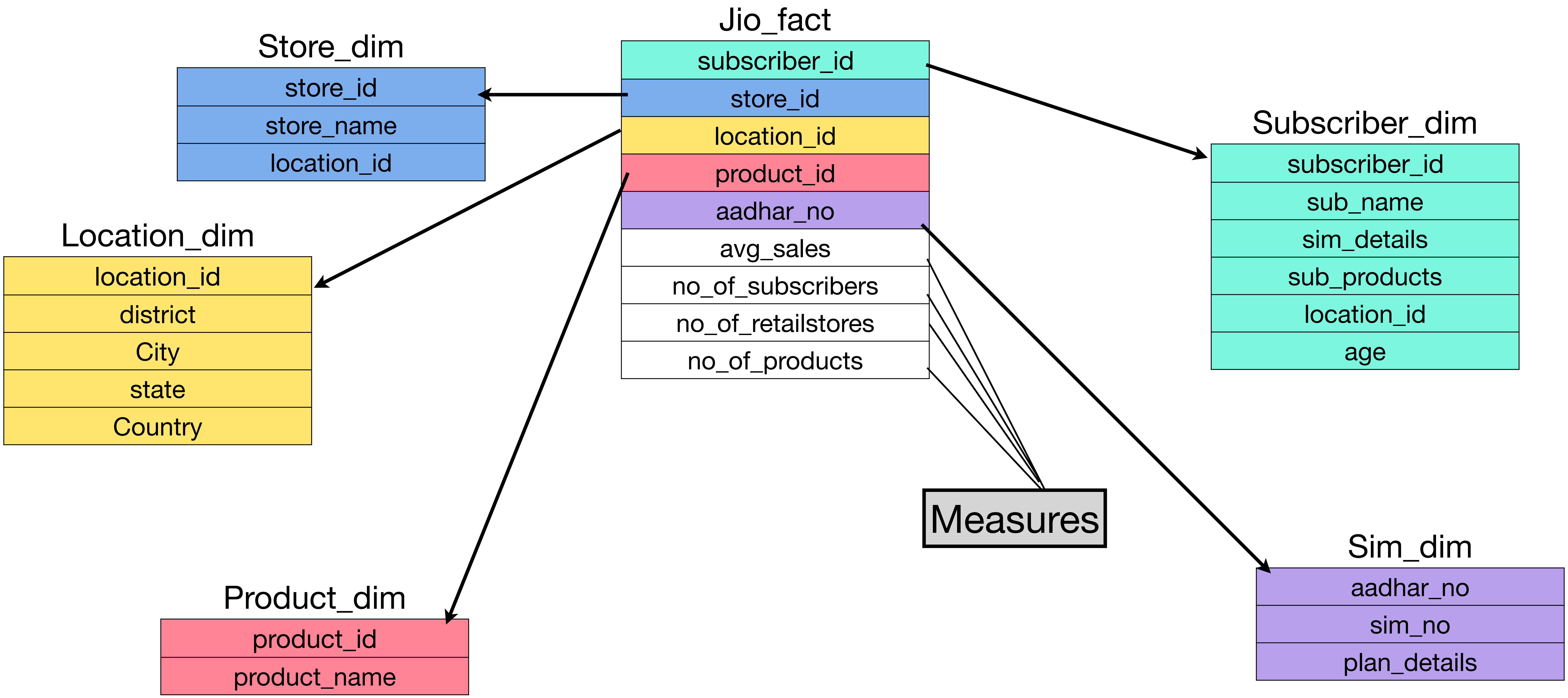
Some **KPI’s of Jio**:

- Average Revenue
- Number of Jio subscribers (about 387.5 million)
- Market Acquisition Rate
  - New users buying Jio
  - Users switching to Jio
- Rate of Growth of Diversified Business (Diverse products & services offered by Jio)
- Number of retail stores (about 11,784)
- Market share acquired by Jio



**Types of keys mentioned in fact constellation

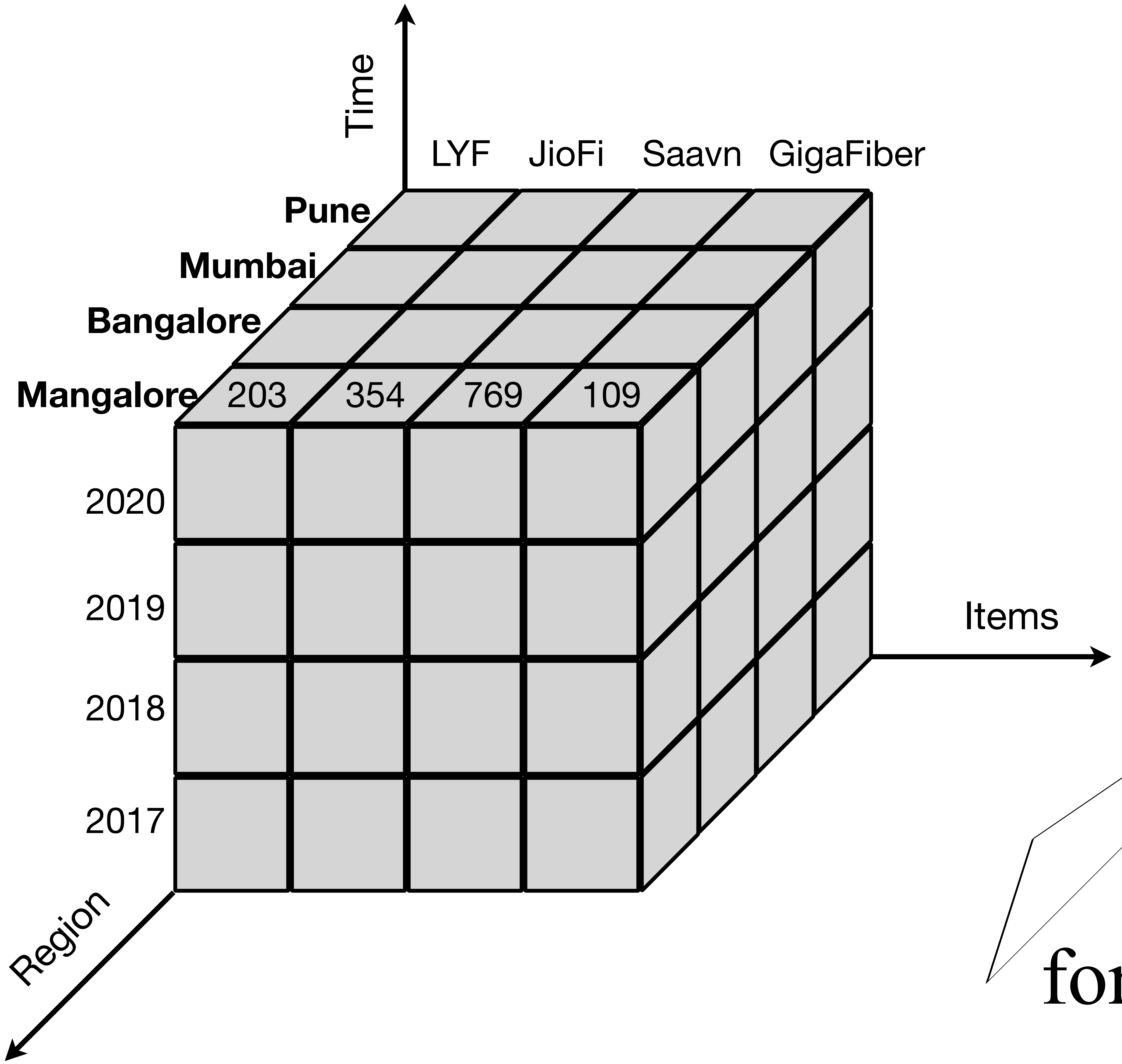
Star Schema



Fact Constellation Schema & Types of Keys

**Since Fact Constellation Schema is an advanced version of Snowflake Schema, I have included Fact constellation directly

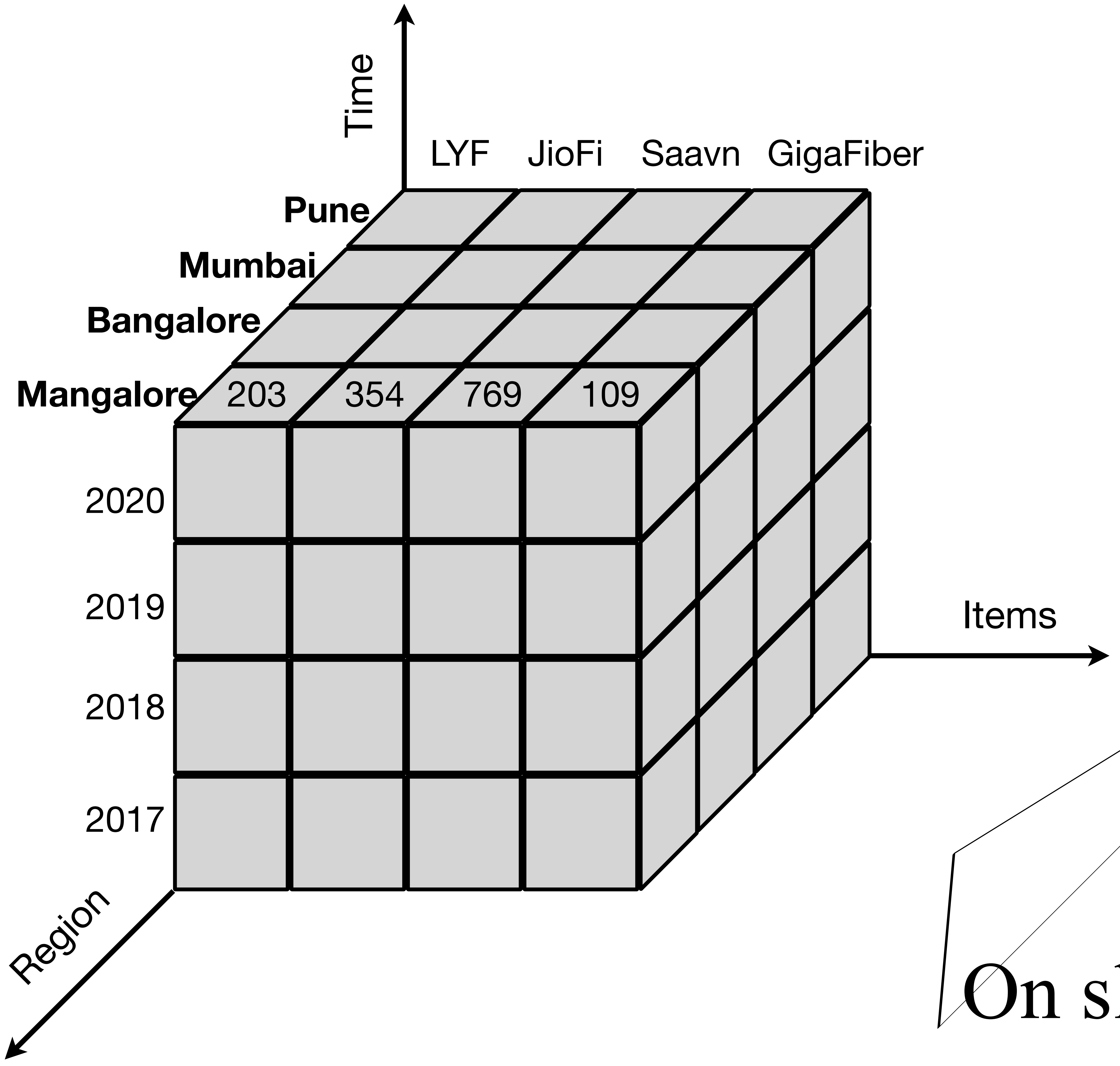
OLAP Operations (On-Line Analytics Processing)



	LYF	JioFi	Saavn	GigaFiber
Pune				
Mumbai				
Bangalore				
Mangalore	203	354	769	109

Slicing
for time = 2020

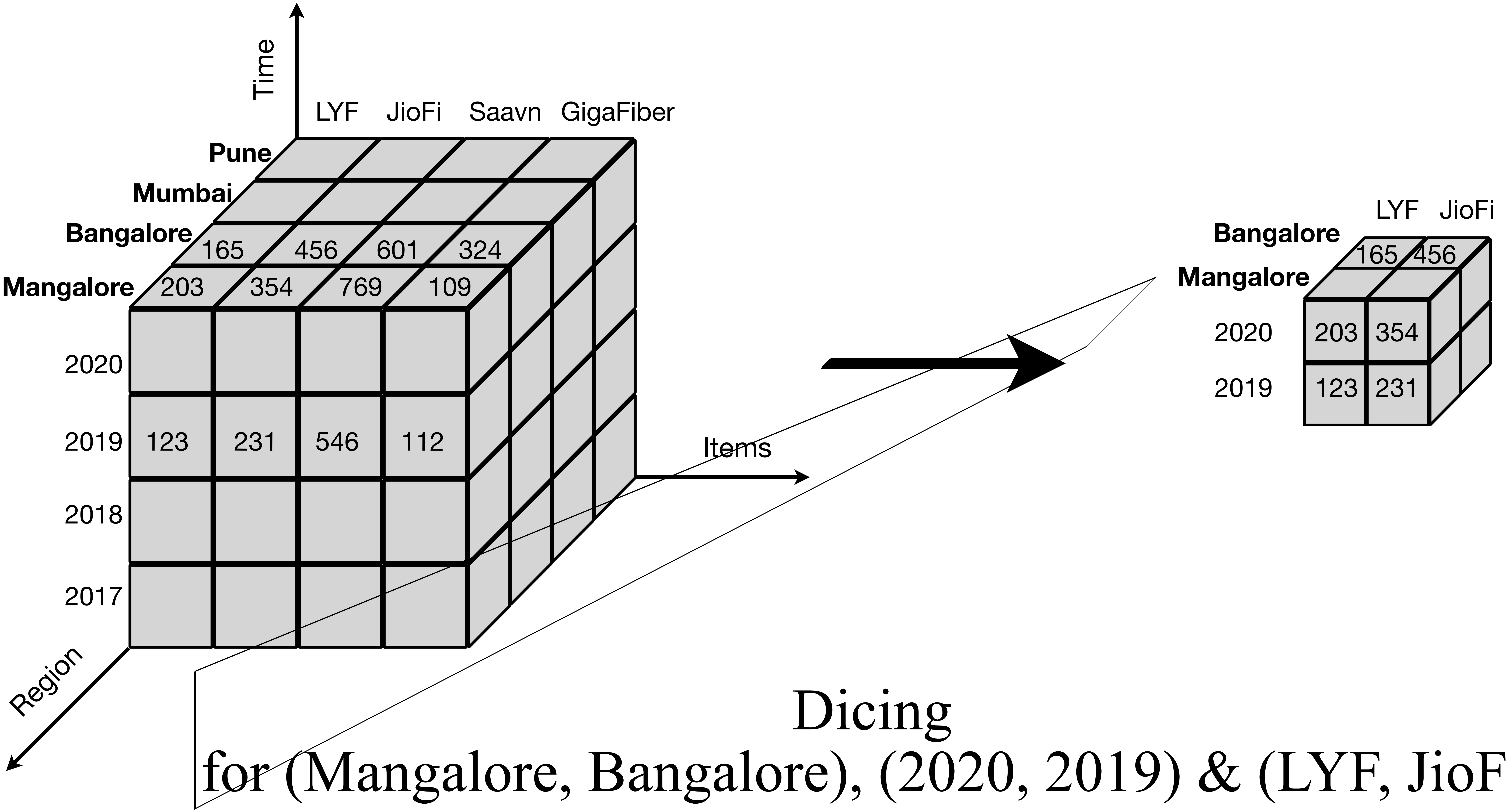
OLAP Operations (On-Line Analytics Processing)



	Pune	Mumbai	Bangalore	Mangalore
LYF				203
JioFi				354
Saavn				769
GigaFiber				109

Pivot
On slicing operation

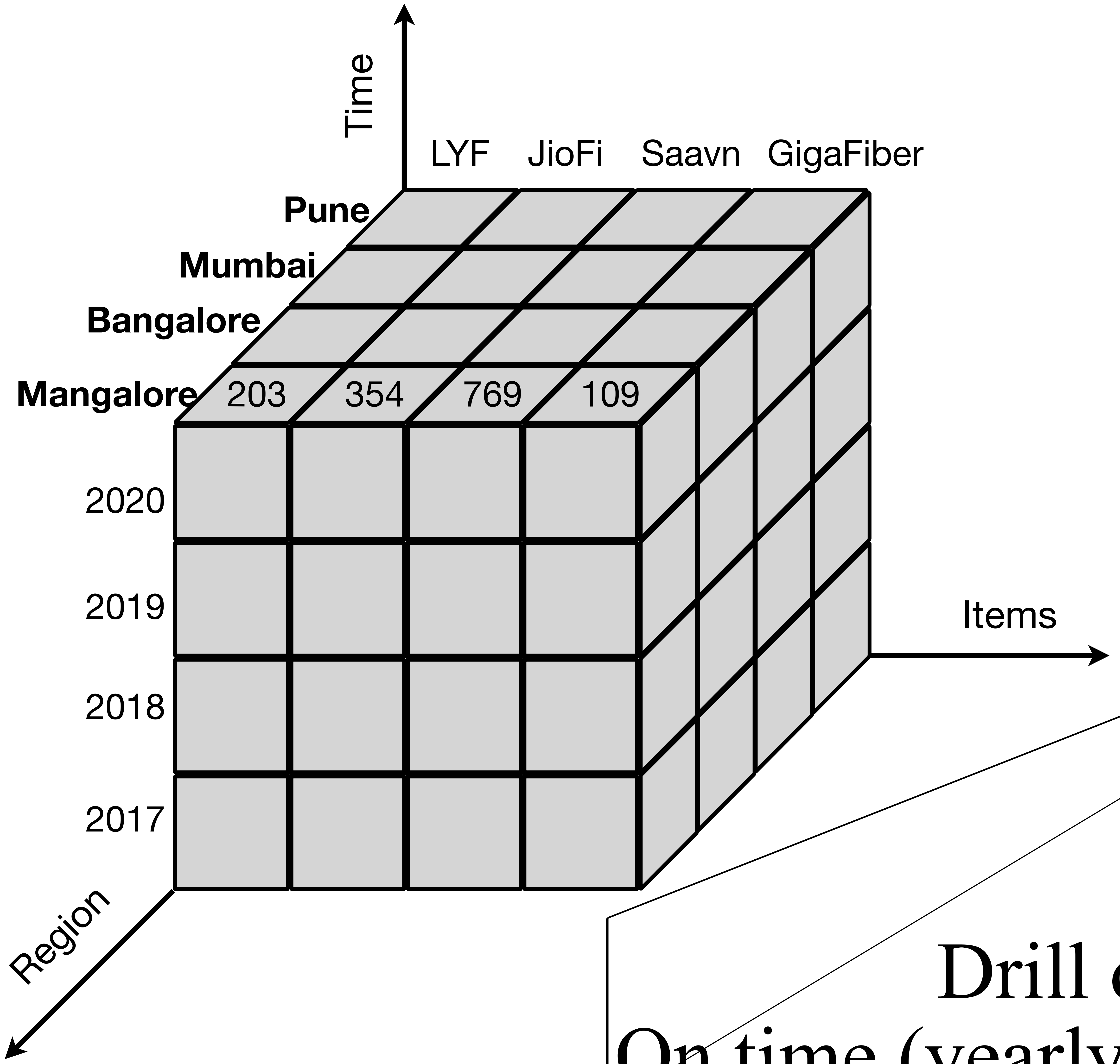
OLAP Operations (On-Line Analytics Processing)

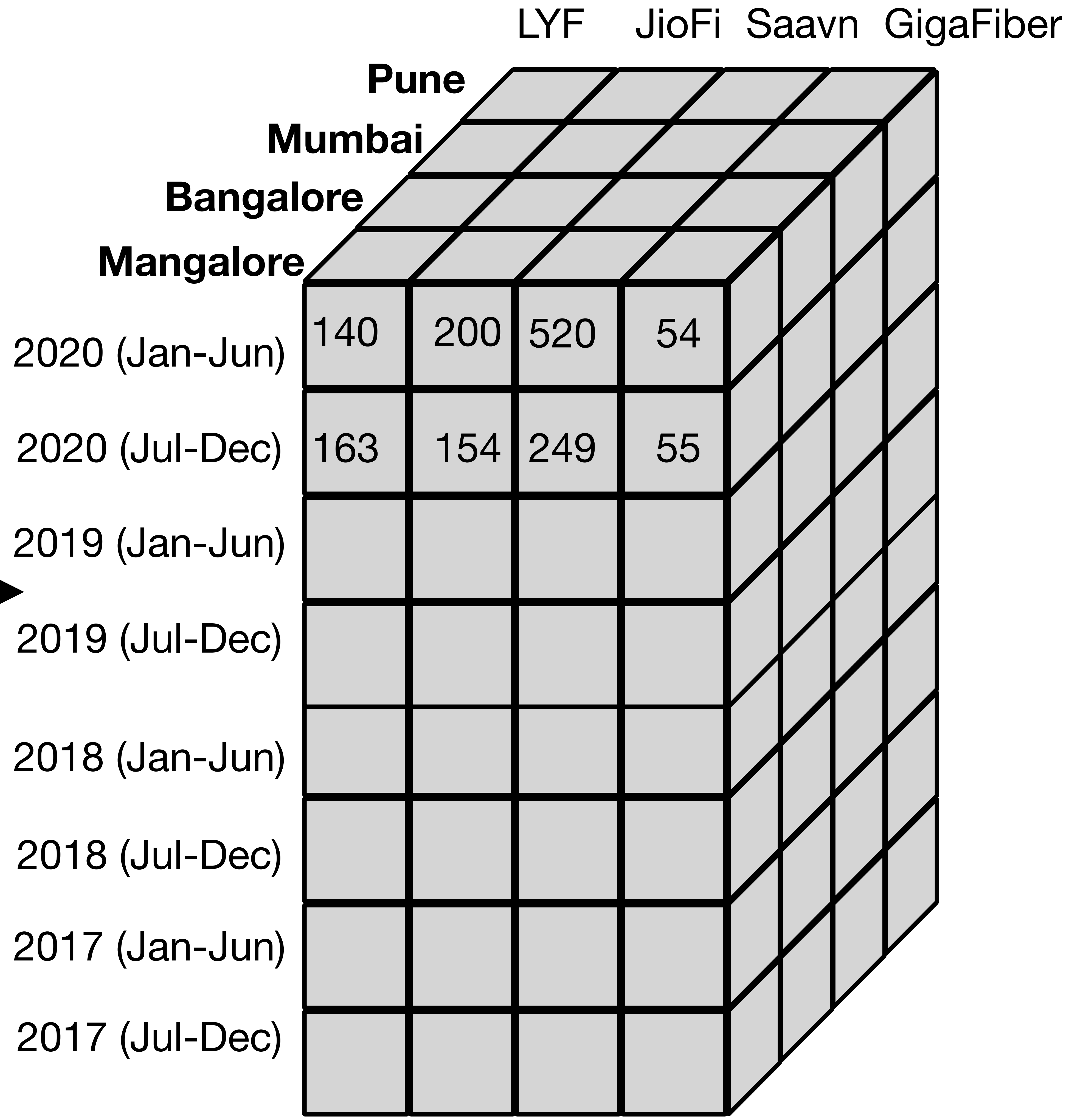


Dicing

for (Mangalore, Bangalore), (2020, 2019) & (LYF, JioFi)

OLAP Operations (On-Line Analytics Processing)

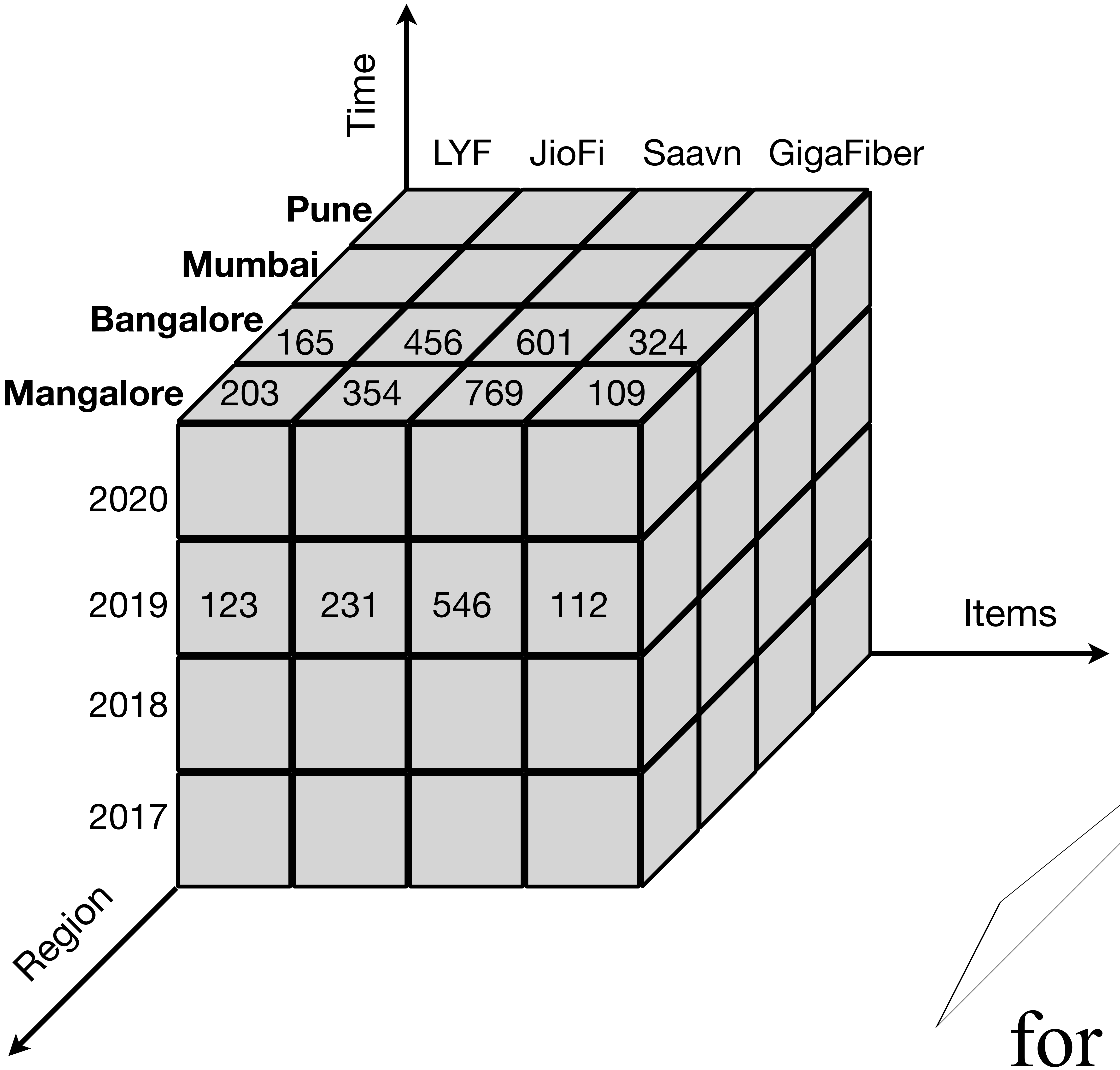


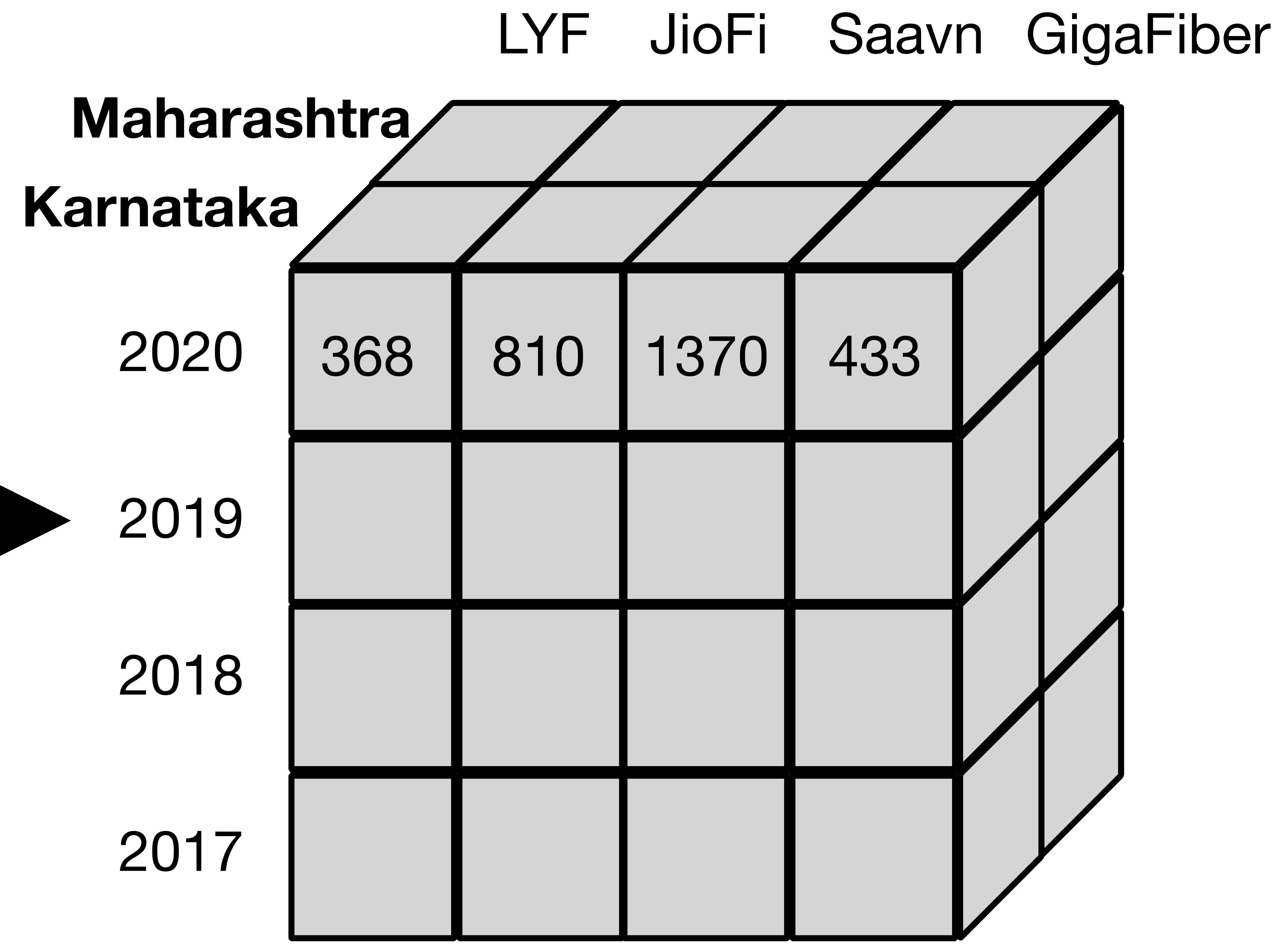


Drill down

On time (yearly to half-yearly)

OLAP Operations (On-Line Analytics Processing)





Roll up
for cities to states

Role of Data Warehousing

Firstly, What is a data warehouse?

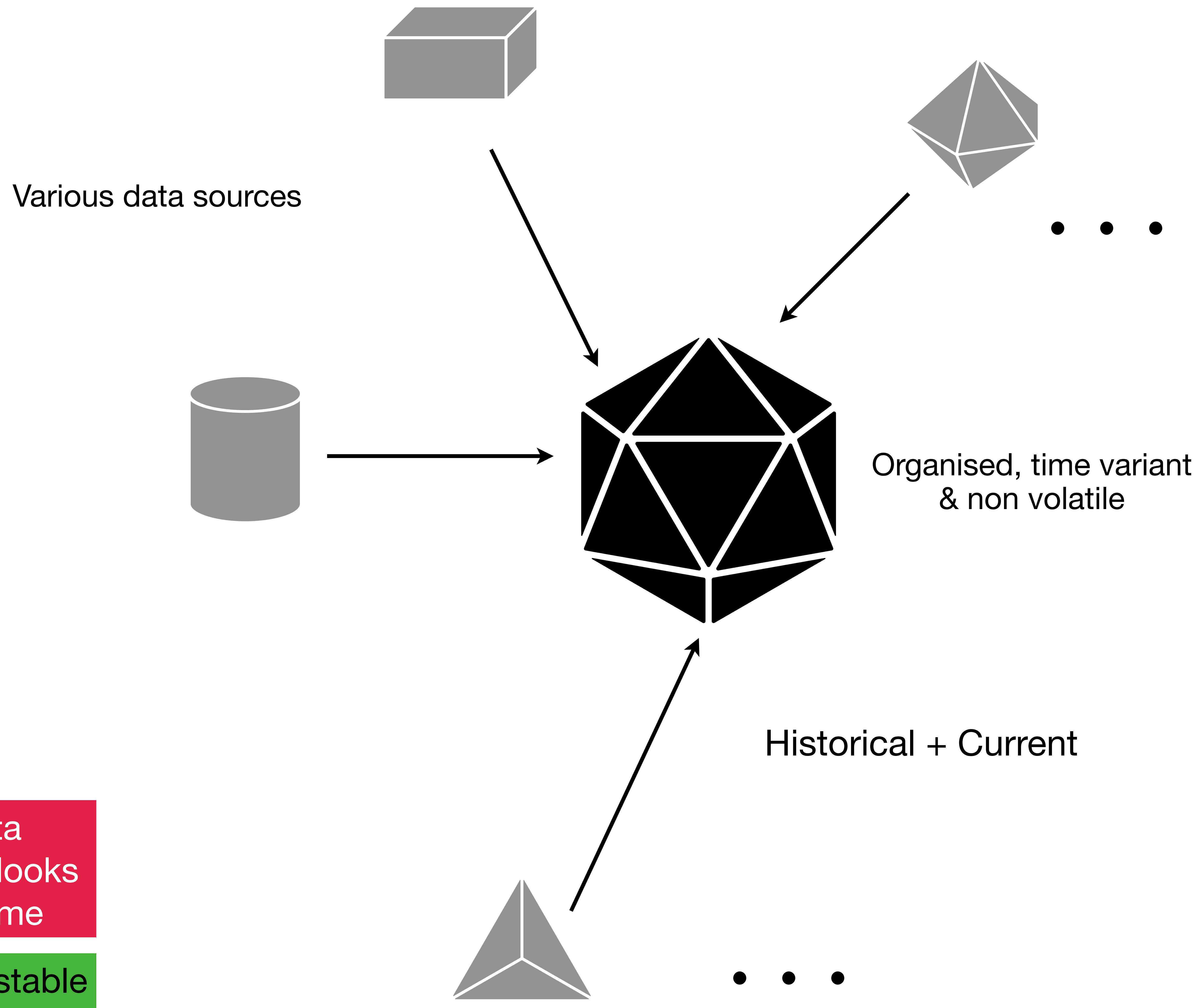
- A data warehouse is a **repository** (archive of an RDB) of information that has been gathered from various different resources and stored at a single site
- It is **subject-oriented, integrated, time variant** and **non volatile**
- It stores **historical** data along with current data trends which are important for businesses to carry out data analysis (OLAP) operations

Subject-oriented: analysis of a particular subject of data

Time-variant: data warehouse analysis looks at changes over time

Integrated: integrating multiple heterogeneous data

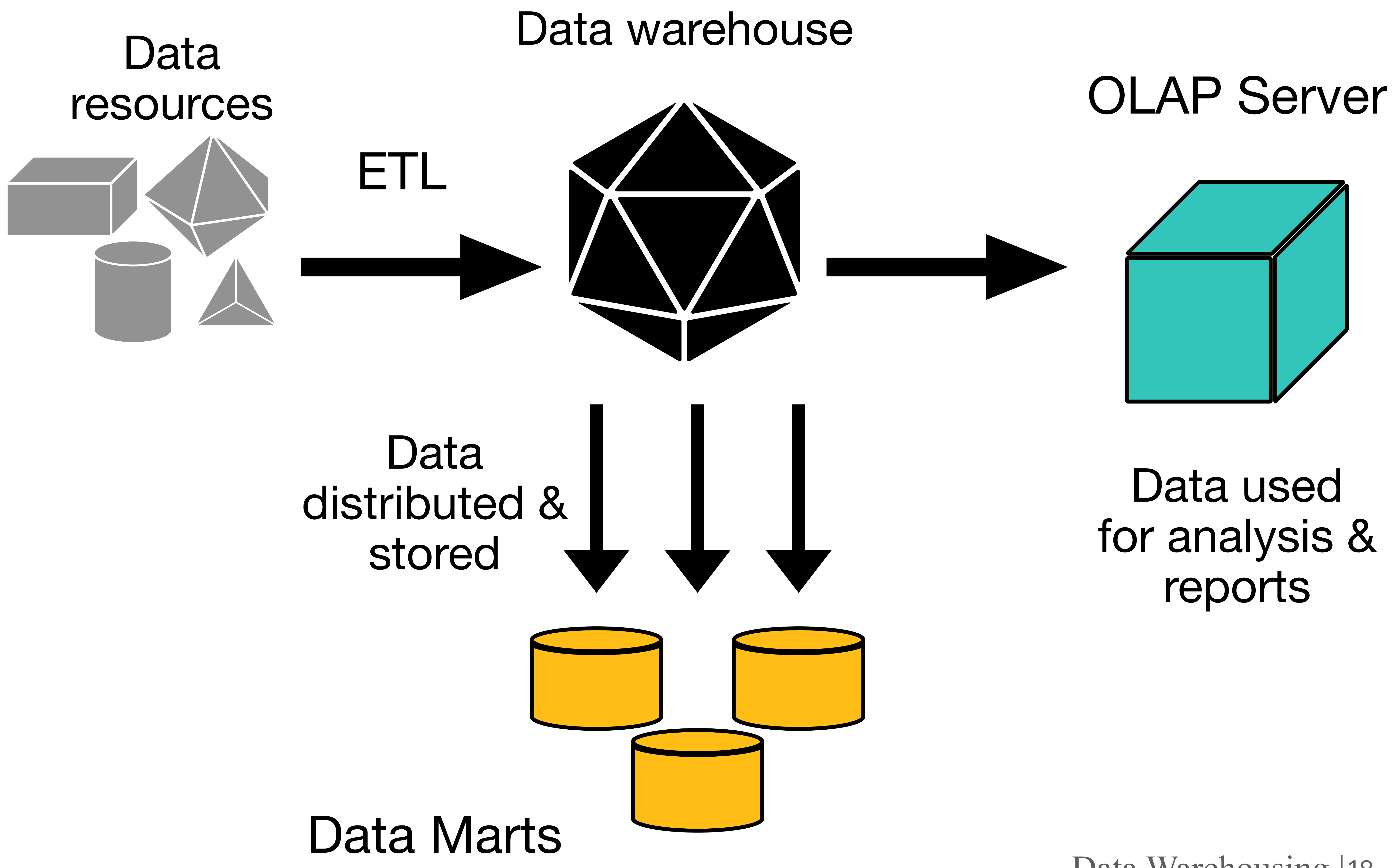
Non-volatile: data is stable & doesn't change



Role of Data Warehousing (contd.)

Now, What is a data warehousing?

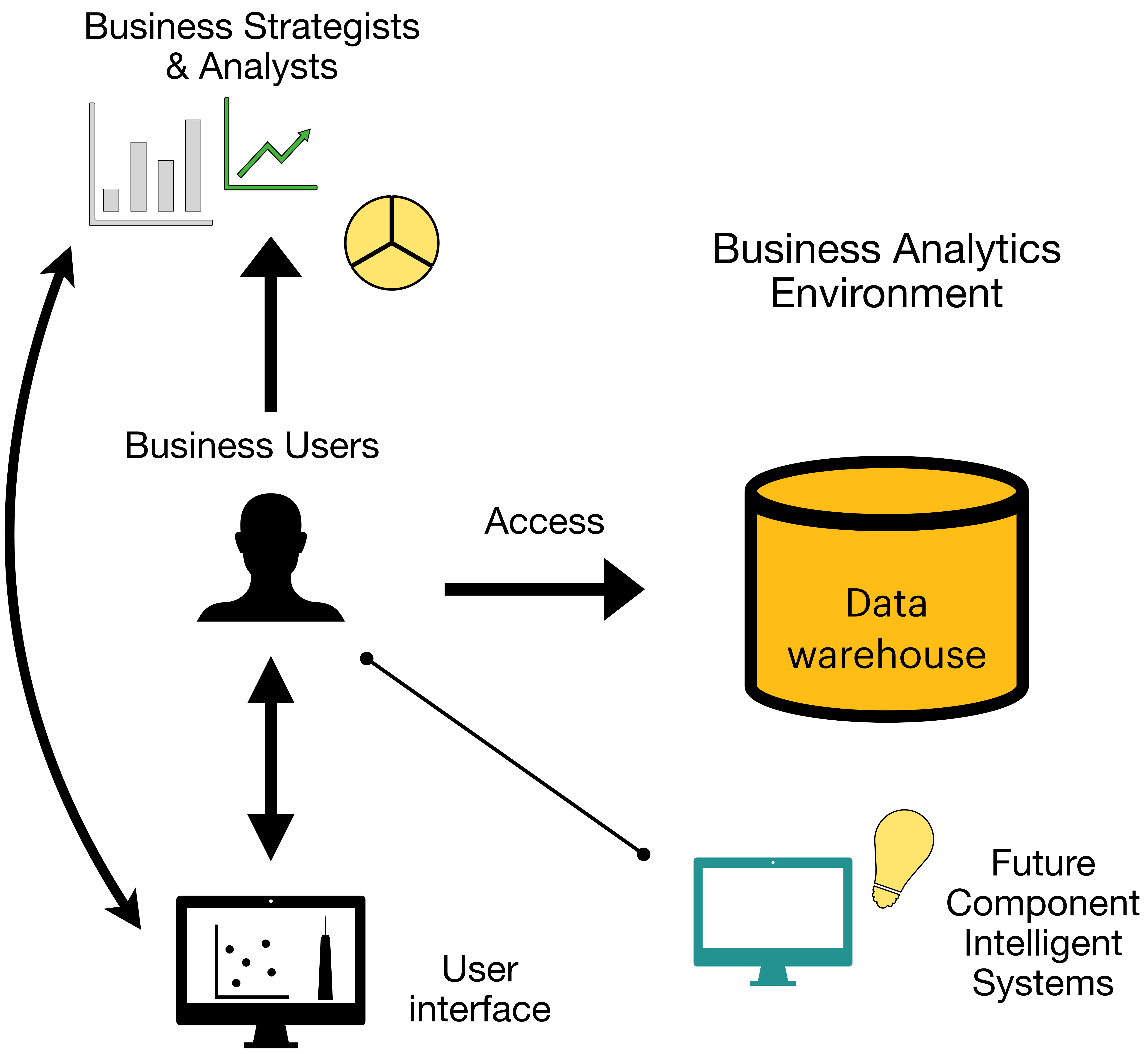
- Data warehousing is the process of taking data from legacy databases (traditional DBMS) and converting it into more **organised formats**
- This conversion happens by a process called **ETL** which is Extract, Transform, Load
- The “organised formats” are the data cubes which can be viewed in **multiple dimensions** & are stored in these warehouses
- These data cubes are modelled into various **schemas** like star schema, snowflake schema fact constellation and galaxy schema
- Data in the warehouses is also distributed & stored in **Data marts** which is basically a subset of the large data and is useful only to certain group of customers



Role of Data Warehousing to Jio

Data warehousing enables

- Analysis of terabytes of multidimensional data for specific products like JioPrime, for personalised commerce & consumer engagement business
- Handling and processing complex multidimensional data, its management & data migration efficiently
- The analytical capabilities of a data warehouse allows Jio to derive valuable insights from their data to improve their decision-making. This is where historical data is helpful. Over time the current data becomes historical data and it can be used to analyse various trends for the business
- Allows efficient data visualisation & presentation using tools like crosstabs, tables, charts & graphs that can be used by business users



Slowly Changing Dimensions (SCD's)

What are SCD's?

- SCD's or Slowly Changing Dimensions are the attributes which basically change slowly over time
- They are of 4 types:
    1. Type 0 - these are passive and never change with time
    2. Type 1 - updating / overwriting of data over time (like packaging type of a cold drink brand), may cause performance issues
    3. Type 2 - aka partitioning history, –Every time a change happens a new surrogate primary key is assigned and all fact tables now onwards use this new foreign key
    4. Type 3 - aka alternative realities, old value of the attribute remains valid as a second choice (for example, current address updated for property purposes)

Ways to handle SCD's

1. Type 1 - **<u>Overwriting new data over old data</u>** - no history is kept; easy to maintain

Cust_id	Cust_name	Cust_type
13246	Aishwarya	Corporate

→

Cust_id	Cust_name	Cust_type
13246	Aishwarya	Retail

2. Type 2 - **<u>Creating new record</u>** - we capture the attribute change by adding a new row with a new surrogate key to the dimension table

Cust_id	Cust_name	Cust_type	Current_ status
13246	Aishwarya	Corporate	N

→

Cust_id	Cust_name	Cust_type	Current_ status
13246	Aishwarya	Retail	Y

Ways to handle SCD's (contd)

3. Type 3 - **Adding a new column** - in this, only the current and previous value of the attribute is kept in the database. For example, the new value of password is in the "current_password" column and old value in "previous_password" column

Cust_id	Cust_name	Current_type	Prev_type
13246	Aishwarya	Corporate	Corporate

→

Cust_id	Cust_name	Current_type	Prev_type
13246	Aishwarya	Retail	Corporate

4. Type 4 - **Using historical table** - a separate historical table is used to track all dimensions' attribute historical changes for each dimension

Cust_id	Cust_name	Cust_type
13246	Aishwarya	Corporate

→

Cust_id	Cust_name	Cust_type	Start_date	End_date
13246	Aishwarya	Corporate	20-10-18	10-7-19
13246	Aishwarya	Retail	11-7-19	22-12-19
13246	Aishwarya	Other	23-12-19	4-5-21

Ways to handle SCD's (contd)

6. Type 6 - **<u>Combine approaches 1 + 2 + 3</u>** (1+2+3=6) - create a new record with surrogate key with an additional column in a different historical table; so in this method, we can handle those 3 types of SCD's in a single methodology

Cust_id	Cust_name	Current_cust _type	Previous_cust _type	Start_date	End_date	Current_status
13246	Aishwarya	Corporate	Corporate	20-10-18	10-7-19	Y
13246	Aishwarya	Retail	Corporate	11-7-19	22-12-19	N
13246	Aishwarya	Other	Retail	23-12-19	4-5-21	Y

Data sources

- Lecture notes for Unit 1, 2, 3
- https://www.jio.com/en-in/apps
- https://iide.co/case-studies/reliance-jio-marketing-strategy/
- https://www.ibm.com/docs/en/ida/9.1?topic=phase-step-identify-grain
- https://www.hirist.com/j/reliance-jio-data-warehousing-lead-data-management-8-11-yrs-291389.html
- https://www.oracle.com/in/database/what-is-a-data-warehouse/
- https://www.datawarehouse4u.info/SCD-Slowly-Changing-Dimensions.html

Thank You!